Graph-informed and FiLM-enhanced Multimodal Fusion for Myocardial Infarction Prediction

Xiantong Xiang, Longxiao Gao, Yuansheng Liu, Ningji Gong, and Yongshun Gong .

***Abstract*— Accurate and timely diagnosis of cardiovascular diseases, particularly myocardial infarction (MI), remains a critical clinical challenge. Existing electrocardiogram (ECG) analysis methods often rely solely on a single data modality, such as raw signals or waveform images, which limits their ability to capture the broader physiological context. To address this limitation, we propose GFM-MIP, a Graph-informed and FiLM-enhanced Multimodal Fusion framework for myocardial infarction prediction. GFM-MIP integrates 12-lead ECG time-series signals, ECG images, and laboratory test results through a unified architecture. Specifically, it employs a Graphormer encoder to model inter-lead dependencies in ECG signals and a Vision Transformer to extract morphological patterns from ECG images, both modulated by patient-specific laboratory features using Feature-wise Linear Modulation (FiLM). A Transformer-based fusion module captures cross-modal interactions, while a contrastive learning objective encourages alignment between signal and image modalities. Experimental results on a real-world clinical dataset and three public benchmarks demonstrate that GFM-MIP consistently outperforms state-of-the-art baselines across multiple evaluation metrics. Ablation studies further validate the contribution of each modality and architectural component. The proposed framework offers a clinically meaningful and scalable solution for robust, multimodal cardiovascular diagnosis.**

***Index Terms*— ECG, Multimodal Learning, Cross-modal Fusion, Myocardial Infarction Diagnosis**

I. Introduction

MYOCARDIAL infarction (MI), a severe manifestation of cardiovascular disease (CVD), remains a major global health threat, causing significant morbidity, mortality, and healthcare expenditures worldwide. Rapid and accurate identification of MI is critically important for initiating timely clinical interventions, which can significantly reduce the risk of adverse cardiac events, improve patient outcomes, and guide effective personalized treatment strategies. Electrocardiography (ECG), particularly the 12-lead ECG, is a frontline diagnostic tool widely adopted in clinical practice, owing to its cost-effectiveness, noninvasiveness, and ease of acquisition even under emergency conditions [1]–[3] . However, despite its prevalent use, accurate interpretation of ECG data continues to pose considerable diagnostic challenges. ECG signals exhibit complex morphological patterns and temporal variations, with subtle pathological manifestations often masked by physiological noise and individual patient variability. The accurate detection of MI requires careful analysis of ECG waveforms, particularly ST-segment deviations, T-wave inversions, and Q-wave abnormalities, which demand substantial expertise and are prone to subjective interpretation errors even among trained clinicians.

Conventional approaches to automated ECG-based MI diagnosis [4]–[12] predominantly rely on single-modality inputs, typically involving raw ECG time-series signals [13] or waveform-derived images [14], coupled with traditional deep learning models such as convolutional neural networks (CNNs) and recurrent neural networks (RNNs). Although these methods have demonstrated progress, they suffer fundamental limitations when applied in clinical environments. First, unimodal approaches inherently restrict the diagnostic model to only one aspect of cardiac physiology, thereby neglecting complementary diagnostic information available from alternative data sources such as biochemical markers or visual morphology. This unimodal focus severely limits the clinical interpretability and robustness of diagnostic predictions, particularly in atypical or borderline clinical presentations. Second, many existing computational models consider ECG leads as independent sequences, disregarding the underlying anatomical and physiological connections between them. Such oversimplified approaches neglect crucial spatial dependencies and inter-lead correlations, impairing the model's capability to identify subtle yet clinically significant diagnostic patterns across different leads. Additionally, most methods do not exploit auxiliary clinical data such as laboratory biomarkers—including troponin, D-dimer, and inflammatory indicators—that could provide critical context and enhance diagnostic sensitivity and specificity in ambiguous cases [13].

These limitations underscore several fundamental challenges that must be overcome to achieve robust automated

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The additional data related to this paper can be downloaded from `https://physionet.org/content/challenge-2021/1.0.3/`. The code for training and predicting are available at `https://github.com/XiangXT-TIME/GFM-MIP`.

MI diagnosis using ECG data. Primarily, there is a significant challenge in effectively integrating diverse physiological signals—including temporal, morphological, and biochemical information—into a cohesive and interpretable diagnostic representation [15], [16]. This integration demands advanced computational strategies that can jointly model structured temporal dependencies, patient-specific physiological variations, and complex interactions between heterogeneous data sources. Another critical challenge lies in accurately modeling the structured relationships between ECG leads. The ECG's intrinsic spatial structure, driven by cardiac anatomy and electrophysiological pathways, provides essential diagnostic context, yet modeling these inter-lead interactions effectively remains under-explored. Furthermore, the significant variability in patient-specific ECG characteristics and systemic physiological states presents a formidable generalization challenge. Effective computational diagnostic systems must be capable of adapting to the unique electrophysiological and clinical context of individual patients, especially when encountering varied patient demographics, different clinical settings, and data quality inconsistencies typically encountered in real-world healthcare environments.

In this work, we propose a unified framework to address these challenges by integrating multimodal physiological data through structured encoding and semantic alignment. We propose GFM-MIP (Graph-informed and FiLM-enhanced Multimodal Fusion for Myocardial Infarction Prediction), a novel deep learning framework that integrates three complementary modalities: ECG time-series, ECG images, and laboratory test results. Specifically, GFM-MIP employs a Graphormer-based encoder [17] to capture the relational structure among ECG leads, and a Vision Transformer to extract morphological features from ECG images. Patient-specific physiological context is injected into both branches via Feature-wise Linear Modulation (FiLM), enabling dynamic modulation of feature extraction. A Transformer-based fusion module aggregates the cross-modal representations, while a symmetric contrastive learning objective aligns time-series and image modalities in the shared latent space. The entire framework is jointly optimized using a hybrid loss that balances supervised classification and contrastive alignment.

Our main contributions are summarised as follows:

- We propose a unified multimodal framework for MI prediction that integrates ECG signals, ECG images, and laboratory biomarkers.
- We introduce a Graphormer-based temporal encoder to model the spatial and temporal dependencies among ECG leads.
- We leverage FiLM-based modulation to inject patient-specific physiological information into both temporal and spatial branches.
- We design a contrastive learning objective to align heterogeneous modalities and improve multimodal consistency.
- We conduct extensive experiments on real-world clinical datasets and public benchmarks, demonstrating that our method achieves state-of-the-art performance across multiple evaluation metrics.

Through these innovations, GFM-MIP addresses key limitations of traditional single- modality ECG approaches, providing a robust, interpretable, and clinically valuable solution for myocardial infarction diagnosis.

II. Related Work

A. ECG-Based Deep Learning Methods

Automated ECG interpretation has been a long-standing research focus in computational cardiology. Early studies primarily relied on handcrafted features and traditional classifiers. With the rise of deep learning, CNN-based and RNN-based architectures became dominant due to their ability to capture local patterns and sequential dynamics from raw ECG signals. Liu et al. [18] proposed a CNN model for detecting acute myocardial infarction (MI), showing improved diagnostic performance over conventional rule-based systems. Jafarian et al. [19] explored both shallow and deep neural networks to localize infarct regions based on signal morphology, while Lee et al. [20] utilized deep learning to detect low ejection fraction from ECG signals.

More recently, transformer-based models have been introduced for ECG analysis due to their superiority in modeling long-range dependencies. Nie et al. [21] demonstrated that temporal self-attention can enhance ECG forecasting performance in long sequences. However, most existing methods are unimodal, focusing exclusively on time-series signals or 2D ECG images. Such approaches overlook the complementary value of biochemical indicators and lack the ability to capture diverse physiological cues, limiting their diagnostic robustness in clinical settings [22], [23].

B. Graph Neural Networks and Transformers in Biomedical Applications

Graph-based learning has become increasingly prominent in medical AI due to its strength in modeling structured and relational data. In ECG applications, the 12-lead signal system can be naturally represented as a graph, where each node corresponds to a lead, and edges reflect anatomical or physiological connectivity. Backhaus et al. [24] employed graph models to analyze myocardial strain, emphasizing the importance of spatial dependencies. Similarly, Alkhodair et al. (2022) leveraged GNNs to detect cardiac abnormalities by modeling spatial relations between ECG leads, significantly improving classification interpretability.

In parallel, Vision Transformers (ViTs) have shown competitive results in medical image analysis, outperforming CNNs in capturing global morphological patterns. Kilimci et al. [14] demonstrated the effectiveness of ViTs in ECG image classification. ViT variants have also been successfully applied in other domains such as pathology and ophthalmology, highlighting their broad applicability in medical vision tasks.

Moreover, hybrid architectures that combine graph neural networks and transformer modules have been explored in multi-organ segmentation [25], cancer subtype classification [26], and neuroimaging analysis [27]. These works support the premise that combining relational inductive biases from GNNs with global attention from transformers can yield more expressive biomedical representations. Yet, to our knowledge, few studies have jointly applied GNNs and transformers to the modeling of ECG data across both temporal and morphological dimensions within a unified diagnostic framework.

C. Multimodal Fusion and Contrastive Learning in Clinical Modeling

Multimodal fusion has emerged as a powerful approach to enhance diagnostic accuracy by leveraging diverse patient information [22], [28]–[30]. Several works combine ECG with clinical metadata or laboratory values to improve prediction. Al-Zaiti et al. [31], [32] demonstrated that integrating ECG signals with structured clinical features significantly improves the identification of occlusion MI. Toprak et al. [13] used machine learning on high-sensitivity cardiac troponin (hs-cTn) to achieve more accurate triage decisions. Sun et al. [33] and Kalmady et al. [34] further highlighted the population-scale utility of ECG models augmented with lab data for mortality prediction and multi-condition screening.

Contrastive learning has also gained traction in biomedical representation learning [35], especially for improving modality alignment and generalization. Wei et al. [36] proposed a bimodal masked autoencoder for ECG that incorporates contrastive losses to encourage consistent representations across domains. More recent frameworks like TimeMAE [37] and TimesURL [38] introduced sophisticated augmentation and pretext strategies for time-series contrastive learning, though they are often limited to single-modality or intra-signal tasks.

Despite these advancements, the combination of structured fusion, patient-conditioned modulation, and cross-modal contrastive alignment remains underexplored in clinical modeling. This motivates the development of unified frameworks—such as our proposed GFM-MIP—that jointly address heterogeneity, personalization, and semantic coherence across ECG time-series, images, and lab features.

III. The Proposed Method

To overcome the limitations of existing ECG-based myocardial infarction (MI) diagnostic approaches, we propose GFM-MIP (Graph-informed and FiLM-enhanced Multimodal Fusion for Myocardial Infarction Prediction), a novel multimodal learning framework that integrates ECG time-series signals, ECG images, and laboratory biomarkers into a unified, clinically interpretable diagnostic representation (Fig 1).

GFM-MIP offers a unified multimodal solution for cardiovascular disease prediction by integrating ECG time-series, ECG images, and laboratory test features. Through the combination of modality-specific encoders, FiLM-based patient-level modulation, and a Transformer-based fusion mechanism, the architecture is capable of capturing complex temporal, morphological, and physiological patterns. By incorporating contrastive learning into the optimization process, GFM-MIP ensures semantic consistency across modalities, encouraging robust and interpretable feature representations. This integrated design not only enhances classification performance but also aligns well with clinical reasoning, thereby advancing the development of explainable and personalized diagnostic models.

Specifically, the proposed GFM-MIP framework comprises four key components:

- **Graphormer-based ECG temporal encoder**, which explicitly models the structured inter-lead dependencies and temporal dynamics inherent in 12-lead ECG signals.
- **Vision Transformer-based morphological encoder**, which extracts global morphological features from ECG waveform images.
- **Feature-wise Linear Modulation (FiLM) mechanism** that dynamically integrates laboratory-derived patient biomarkers into both temporal and morphological encoders, enabling patient-specific feature modulation.
- **Transformer-based cross-modal fusion module** that combines modality-specific representations to capture complex cross-modal interactions, producing a unified latent diagnostic representation.

A distinctive innovation of GFM-MIP is the explicit incorporation of laboratory features ($\mathbf{x}_{\text{lab}}$), which are notably absent from most publicly available ECG datasets. Incorporating lab-derived features allows patient-specific modulation of ECG representations through FiLM layers, greatly enhancing the specificity and adaptability of the model. For instance, patients exhibiting similar ECG waveforms can be effectively differentiated by their distinct troponin levels, which GFM-MIP leverages to tailor representation encoding accordingly.

Importantly, the FiLM mechanism provides a powerful means for integrating sparse yet clinically significant biomarkers, allowing the model to adaptively reweight ECG-derived features based on patient-specific biochemical profiles, which has not been fully leveraged by existing ECG frameworks.

To further enhance semantic coherence between the modalities, a cross-modal contrastive learning objective is employed. This objective explicitly aligns the representations learned from the ECG time-series and image branches, thereby promoting robust and consistent multimodal representations. Finally, the framework is optimized using a joint loss function, combining supervised classification and contrastive alignment, thus ensuring diagnostic accuracy and representational consistency.

In the subsequent sections, we detail each of these components systematically, clearly illustrating their design rationale, mathematical formulation, and implementation.

A. Multimodal Inputs and Problem Formulation

Traditional ECG classification models predominantly rely on a single modality, such as either time-series signals or waveform images, and typically overlook additional clinical insights from complementary sources like laboratory test results. GFM-MIP overcomes these limitations by integrating three distinct yet complementary modalities, creating a comprehensive representation of patient health:

$$\mathcal{D}_i = \{\mathbf{X}_{\text{ECG},i}, \mathbf{X}_{\text{img},i}, \mathbf{x}_{\text{lab},i}, y_i\}, \tag{1}$$

where $\mathbf{X}_{\text{ECG},i} \in \mathbb{R}^{12\times T}$ denotes the 12-lead ECG time series, $\mathbf{X}_{\text{img},i} \in \mathbb{R}^{1\times 224\times 224}$ represents the ECG grayscale image, $\mathbf{x}_{\text{lab},i} \in \mathbb{R}^{D_{\text{lab}}}$ is the laboratory feature vector, and $y_i \in \{0,1\}$ is the binary diagnostic label.

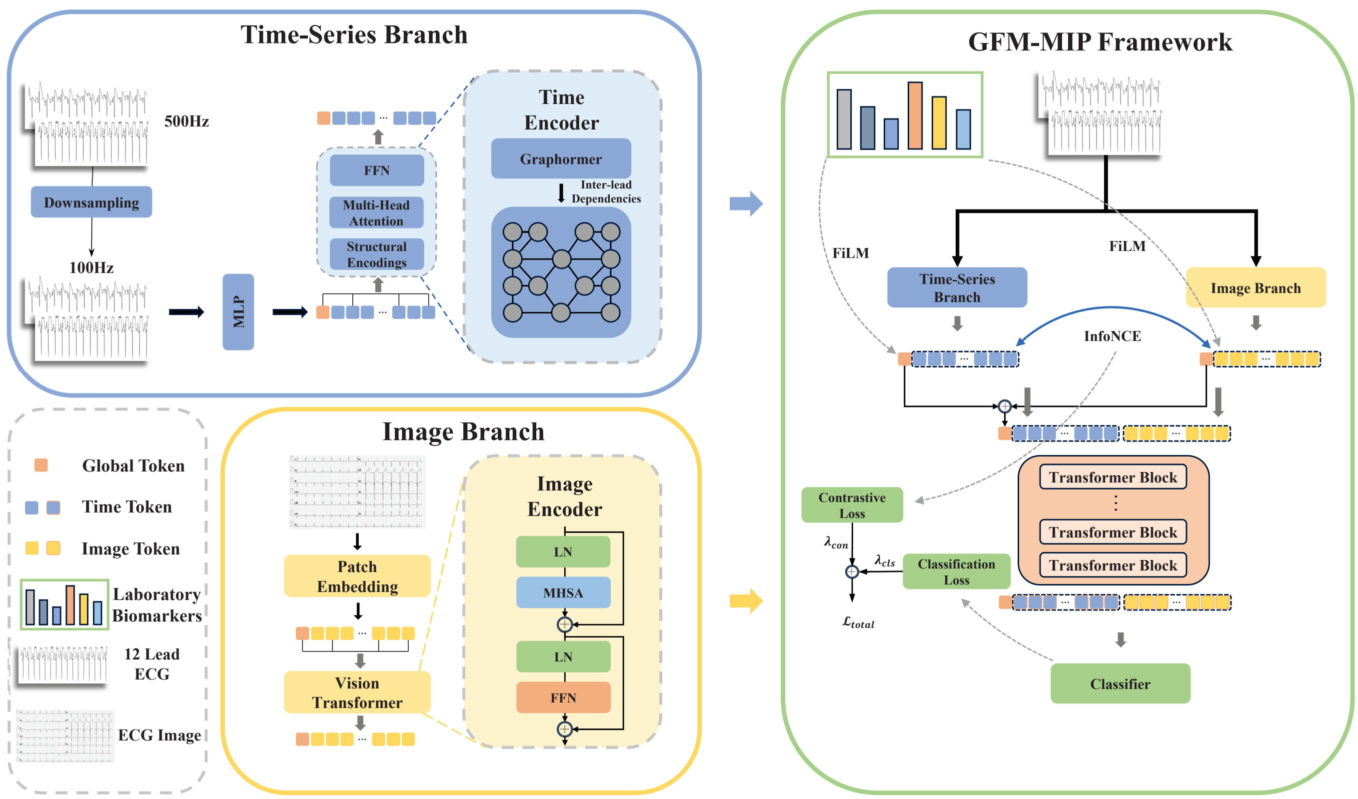

Fig. 1. Overview of the GFM-MIP framework. The proposed model performs multimodal myocardial infarction classification by jointly leveraging ECG time-series, ECG images, and laboratory biomarkers. A Graphormer encoder extracts inter-lead temporal features from ECG signals, while a Vision Transformer captures morphological patterns from ECG images. Both branches are dynamically modulated by laboratory features via FiLM layers to enable patient-specific encoding. The representations are then integrated through a Transformer-based fusion module that captures fine-grained cross-modal interactions. A contrastive learning objective further aligns the modalities in a shared latent space, enhancing robustness and diagnostic accuracy.

The integration of these modalities is motivated by their inherent diagnostic complementarity and physiological interpretability in cardiovascular medicine:

ECG Time Series ($\mathbf{X}_{\mathrm{ECG}}$): Provide high-resolution temporal information reflecting cardiac electrophysiological activity. Each ECG lead offers distinct anatomical perspectives, enabling precise identification of abnormalities like ST-segment elevation, T-wave inversion, and arrhythmias. For example, characteristic ST-segment elevation in leads V2–V4 typically signals anterior myocardial infarction.

ECG Images ($\mathbf{X}_{\mathrm{img}}$): Capture a two-dimensional representation of ECG waveform morphology, preserving waveform shapes, amplitude relationships, and rhythm patterns in alignment with clinical visual interpretation practices. This visual modality enriches the model with structural and morphological insights not fully accessible through temporal signals alone.

Laboratory Features ($\mathbf{x}_{\mathrm{lab}}$): Offer essential systemic physiological context through biomarkers like troponin (cardiac injury indicator), D-dimer (marker of thrombosis), and white blood cell count (inflammation marker). These features provide critical supplementary clinical information, facilitating differential diagnosis even in scenarios presenting similar ECG patterns.

By integrating these three modalities, GFM-MIP significantly enhances representation robustness. Temporal ECG signals encapsulate detailed electrophysiological dynamics, images deliver morphological context, and laboratory data add systemic physiological dimensions. The integration of lab biomarkers significantly enhances our model's clinical relevance and predictive sensitivity, particularly in differentiating ECG waveforms that may appear similar but reflect distinct underlying physiological states. This multimodal strategy mitigates common challenges such as signal noise, incomplete data, and ambiguous waveform interpretation often encountered with unimodal methods.

To operationalize this multimodal design, we developed a custom dataset featuring synchronized acquisition of all three modalities. This dataset facilitates precise alignment and integration of temporal, spatial, and biochemical information, marking both conceptual and practical advancements over existing unimodal and bimodal datasets.

Collectively, GFM-MIP synthesizes cardiac electrophysiological data ($\mathbf{X}_{\mathrm{ECG}}$), waveform morphology ($\mathbf{X}_{\mathrm{img}}$), and systemic physiological status ($\mathbf{x}_{\mathrm{lab}}$) into a unified and clinically aligned diagnostic representation ($\mathcal{D}_i$). This comprehensive approach significantly enhances prediction accuracy, model robustness, and clinical interpretability, particularly in complex cardiovascular classification scenarios.

B. Graphormer-based ECG Temporal Encoding with Laboratory-guided FiLM Modulation

To model the structured spatiotemporal nature of 12-lead ECG signals, GFM-MIP employs a Graphormer-based encoder augmented with Feature-wise Linear Modulation (FiLM). This design captures both inter-lead dependencies and patient-specific contextual variation.

Let $\mathbf{X}_{\text{ECG}} = [\mathbf{x}_1, \ldots, \mathbf{x}_{12}] \in \mathbb{R}^{12\times T}$, where each $\mathbf{x}_\ell \in \mathbb{R}^T$ represents the time series from lead ℓ. Each lead is projected to a latent representation using a learnable linear transformation:

$$\mathbf{h}_\ell^{(0)} = \mathbf{W}_{\text{proj}} \cdot \mathbf{x}_\ell + \mathbf{b}_{\text{proj}} \in \mathbb{R}^d, \quad \forall \ell \in \{1, \ldots, 12\}. \tag{2}$$

We then encode lead-wise spatial relationships via a Graphormer block, in which spatial biases are implicitly learned and incorporated into the attention mechanism. Specifically, the encoded node representations $\mathbf{H}^{(l)} \in \mathbb{R}^{12\times d}$ are updated layer-wise using a graph-based self-attention operator:

$$\text{Attn}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^\top + \mathbf{B}}{\sqrt{d}}\right)\mathbf{V}, \tag{3}$$

where $\mathbf{B} \in \mathbb{R}^{12\times 12}$ represents learnable spatial bias terms among the ECG leads.

To enable context-aware encoding, we apply FiLM modulation using laboratory features $\mathbf{x}_{\text{lab}}$. The FiLM generator maps lab vectors to scale and shift parameters:

$$\gamma = f_\gamma(\mathbf{x}_{\text{lab}}), \quad \beta = f_\beta(\mathbf{x}_{\text{lab}}), \tag{4}$$

which modulate intermediate features at each encoding layer:

$$\mathbf{H}_{\text{FiLM}}^{(l)} = \gamma \odot \mathbf{H}^{(l)} + \beta. \tag{5}$$

This mechanism injects personalized physiological information into the lead representations, enhancing patient-specific discrimination.

Finally, we extract a global token $\mathbf{z}_{\text{ts,cls}}$ by prepending a learnable [CLS] vector and applying an output projection:

$$\mathbf{z}_{\text{ts,cls}} = \text{TransformerEncoder}([\mathbf{h}_{\text{cls}}; \mathbf{H}^{(L)}]) \in \mathbb{R}^d. \tag{6}$$

This component outputs temporally and spatially enriched ECG embeddings, personalized by clinical laboratory context, for downstream fusion and classification.

C. Vision Transformer-based Morphological Encoding Enhanced by Clinical FiLM Modulation

GFM-MIP employs a Vision Transformer (ViT) architecture augmented by Feature-wise Linear Modulation (FiLM) to effectively capture morphological information from ECG images, integrating patient-specific clinical context derived from laboratory biomarkers.

Given an ECG grayscale image $\mathbf{X}_{\text{img}} \in \mathbb{R}^{1\times H\times W}$, we partition it into N patches, each of size $P \times P$. Each patch is flattened and linearly projected into a latent embedding space:

$$\mathbf{z}_i^{(0)} = \mathbf{W}_{\text{patch}} \cdot \text{Flatten}(\mathbf{X}_i) + \mathbf{b}_{\text{patch}} \in \mathbb{R}^d, \quad i = 1, \ldots, N. \tag{7}$$

We prepend a learnable classification token $\mathbf{z}_{\text{cls}}$ to these embeddings and incorporate positional encodings $\mathbf{E}_{\text{pos}}$ to retain spatial context:

$$\mathbf{Z}^{(0)} = [\mathbf{z}_{\text{cls}}; \mathbf{z}_1^{(0)} + \mathbf{e}_1; \ldots; \mathbf{z}_N^{(0)} + \mathbf{e}_N] \in \mathbb{R}^{(N+1)\times d}. \tag{8}$$

To condition these representations on patient-specific clinical information, FiLM modulation parameters $\gamma^{(l)}$ and $\beta^{(l)}$ are computed from laboratory feature vectors $\mathbf{x}_{\text{lab}}$. These parameters modulate embeddings after each transformer encoder block:

$$\mathbf{Z}_{\text{FiLM}}^{(l)} = \gamma^{(l)} \odot \mathbf{Z}^{(l)} + \beta^{(l)}. \tag{9}$$

This modulation injects patient-specific physiological bias into the visual encoding of ECG morphology, facilitating personalized feature extraction.

The final image representation $\mathbf{z}_{\text{img,cls}}$ is obtained from the output [CLS] token:

$$\mathbf{z}_{\text{img,cls}} = \mathbf{Z}_0^{(L)} \in \mathbb{R}^d. \tag{10}$$

Consequently, this branch effectively captures spatially rich morphological features from ECG images, contextualized by laboratory-derived biomarkers, significantly enhancing the multimodal representational power of GFM-MIP.

D. Cross-modal Representation Fusion

To effectively integrate the representations derived from the time-series and image branches, GFM-MIP introduces a dedicated Transformer encoder-based fusion module. This component is designed to capture both modality-specific and cross-modal interactions, encompassing global diagnostic patterns and fine-grained temporal-spatial dependencies. Our Transformer-based fusion module explicitly captures complex interdependencies across and within modalities, significantly outperforming simpler concatenation strategies or shallow fusion layers, as evidenced by our ablation studies.

The two global tokens $\mathbf{z}_{\text{ts,cls}}$ and $\mathbf{z}_{\text{img,cls}}$ are concatenated and projected through a learnable linear transformation to produce a unified multimodal token:

$$\mathbf{z}_{\text{global}} = \mathbf{W}_f[\mathbf{z}_{\text{ts,cls}}; \mathbf{z}_{\text{img,cls}}] + \mathbf{b}_f \in \mathbb{R}^d, \tag{11}$$

where $\mathbf{W}_f \in \mathbb{R}^{d\times 2d}$ and $\mathbf{b}_f \in \mathbb{R}^d$ are trainable parameters. This global token represents a synthesized summary of temporal and morphological information across modalities.

The fused token is then concatenated with the full sequence of lead- and patch-level embeddings:

$$\mathbf{Z}_{\text{fusion}}^{(0)} = [\mathbf{z}_{\text{global}}; \mathbf{H}_{\text{ts}}; \mathbf{H}_{\text{img}}] \in \mathbb{R}^{(1+12+N)\times d}. \tag{12}$$

This input sequence is fed into a stack of Transformer encoder layers that jointly model hierarchical dependencies and semantic interactions across modalities. These layers are responsible for learning attention-driven relationships that may span time-series leads, image patches, or combinations thereof:

$$\mathbf{Z}_{\text{fusion}}^{(L)} = \text{TransformerEncoder}(\mathbf{Z}_{\text{fusion}}^{(0)}). \quad (13)$$

The output at the first position of the final layer—corresponding to the fused global token—is extracted as the comprehensive multimodal representation:

$$\mathbf{z}_{\text{fusion,cls}} = \mathbf{Z}_{\text{fusion}}^{(L)}[0] \in \mathbb{R}^d. \quad (14)$$

This fused embedding encapsulates the full temporal, spatial, and clinical context, serving as the primary input for subsequent classification and contrastive learning modules.

E. Symmetric Cross-modal Contrastive Learning for Representation Alignment

To promote consistent and semantically aligned representations across modalities, GFM-MIP incorporates a contrastive learning mechanism tailored for the multimodal ECG setting. This strategy enhances the coherence of the learned feature space by bringing together embeddings from different modalities that correspond to the same patient, while simultaneously pushing apart those from different patients.

Let $\mathbf{z}_{\text{ts},i}$ and $\mathbf{z}_{\text{img},i}$ denote the global classification token embeddings extracted from the time-series and image branches, respectively, for the i-th patient in a mini-batch of size B. The alignment objective is based on a symmetric InfoNCE loss, formulated as:

$$\mathcal{L}_{\text{con}} = -\frac{1}{2B}\sum_{i=1}^{B}\left(\ell_i^{\text{ts}\to\text{img}} + \ell_i^{\text{img}\to\text{ts}}\right) \quad (15)$$

where

$$\ell_i^{\text{ts}\to\text{img}} = \log\frac{\exp\left(\text{sim}(\mathbf{z}_{\text{ts},i}, \mathbf{z}_{\text{img},i})/\tau\right)}{\sum_{j=1}^{B}\exp\left(\text{sim}(\mathbf{z}_{\text{ts},i}, \mathbf{z}_{\text{img},j})/\tau\right)} \quad (16)$$

$$\ell_i^{\text{img}\to\text{ts}} = \log\frac{\exp\left(\text{sim}(\mathbf{z}_{\text{img},i}, \mathbf{z}_{\text{ts},i})/\tau\right)}{\sum_{j=1}^{B}\exp\left(\text{sim}(\mathbf{z}_{\text{img},i}, \mathbf{z}_{\text{ts},j})/\tau\right)} \quad (17)$$

where $\text{sim}(\cdot,\cdot)$ denotes cosine similarity, and τ is a temperature scaling factor that controls the sharpness of the distribution.

This bidirectional formulation ensures mutual agreement between the representations from both modalities. By minimizing this loss, the model learns to align ECG signal and image representations at the patient level, reinforcing semantic correspondence across heterogeneous views. This contrastive signal acts as a regularization force, complementing the supervised classification objective and contributing to more robust multimodal feature learning under clinical supervision.

F. Joint Optimization with Hybrid Classification and Contrastive Objectives

Final classification in GFM-MIP is performed using the fused multimodal representation $\mathbf{z}_{\text{fusion,cls}}$, which captures global information integrated across time-series, image, and laboratory modalities. This embedding is passed through a linear classifier to produce class logits:

$$\hat{\mathbf{y}} = \text{softmax}(\mathbf{W}_{\text{cls}} \cdot \mathbf{z}_{\text{fusion,cls}} + \mathbf{b}_{\text{cls}}), \quad (18)$$

where $\mathbf{W}_{\text{cls}}$ and $\mathbf{b}_{\text{cls}}$ are learnable parameters. The classification loss $\mathcal{L}_{\text{cls}}$ is computed using the cross-entropy criterion, with optional weighting schemes to address potential class imbalance.

In addition to classification, GFM-MIP incorporates the contrastive loss $\mathcal{L}_{\text{con}}$ previously described in Section D, which aligns modality-specific embeddings in the latent space. This dual-objective training strategy allows the model to simultaneously optimize for predictive accuracy and representational coherence.

The overall training objective combines both components as follows:

$$\mathcal{L}_{\text{total}} = \lambda_{\text{cls}} \cdot \mathcal{L}_{\text{cls}} + \lambda_{\text{con}} \cdot \mathcal{L}_{\text{con}}, \quad (19)$$

where λ_{cls} and λ_{con} are hyperparameters that balance the classification and contrastive learning contributions. This composite loss ensures that GFM-MIP develops both discriminative and well-aligned multimodal representations tailored for robust clinical prediction.

G. Implementation Details

Our model is implemented using PyTorch and trained on a single RTX 4090 GPU. During training, we set the batch size to 32 and employ 4 worker threads for data loading. The total number of training epochs is 50, with the initial learning rate set to 3.26×10^{-5}. A linear warm-up strategy is applied over approximately 13% of the total training steps , followed by cosine annealing for learning rate decay.

We optimize the model using the AdamW optimizer with default parameters and apply a loss function composed of two terms: a supervised classification loss and a contrastive alignment loss. The weighting coefficients for the classification and contrastive objectives are set to $\lambda_{\text{cls}} = 1.0$ and $\lambda_{\text{contrast}} = 0.445$, respectively. The temperature parameter in the contrastive loss is fixed at 0.1.The fusion module adopts a Transformer encoder structure with 3 layers, each consisting of 4 attention heads . The shared fusion representation has a dimensionality of 256.

To address the issue of label imbalance present in the clinical dataset, we apply class weighting during the training process to stabilize optimization and improve the reliability of evaluation metrics.All baseline methods are trained using the recommended hyperparameter settings provided in their original implementations, ensuring a fair comparison with our model.

IV. Results

A. Datasets

To assess the effectiveness of our proposed model, we conducted comprehensive experiments using three publicly available datasets shown in Table I: the Ningbo dataset [39] from Ningbo First Hospital, the PTB-XL dataset [40] provided by Physikalisch-Technische Bundesanstalt, and the Chapman dataset [41] jointly developed by Chapman University and Shaoxing People's Hospital. These datasets consist of 34,905, 21,837, and 10,247 clinical records, respectively, each comprising 12-lead ECG signals spanning 10 seconds. The raw ECG data were sourced from the PhysioNet Computing in Cardiology (CinC) 2021 Challenge database [42], in which all recordings were uniformly sampled at 500 Hz.

In the CinC 2021 database, each ECG record is annotated with one or multiple labels corresponding to various cardiac rhythm categories, mapped to SNOMED-CT codes. Therefore, the experiments conducted on these public datasets involved multi-label classification tasks.

Moreover, to further validate our model's utility in clinical practice, we collaborated with the Second Hospital of Shandong University to construct a new, proprietary dataset comprising 2,123 clinical records. Each record in this dataset includes a 12-lead ECG and corresponding laboratory test results, annotated with a single binary label. Consequently, this particular dataset was employed for binary classification tasks.

B. Ethics Approval

All procedures performed involving human participants were in accordance with the ethical standards of the institutional and/or national research committee and with the 1975 Declaration of Helsinki and its later amendments or comparable ethical standards. This study protocol was approved by the Ethics Committee of the second hospital of Shandong university (Approval Number: KYLL-2023-455). This study was retrospective and all data were anonymized, so the requirement for informed consent from patients could be waived.

C. Metrics

We utilized four widely adopted performance metrics to rigorously evaluate classification performance: accuracy (ACC), sample-level F1 score (F1), area under the receiver operating characteristic curve (AUROC), and area under the precision-recall curve (AUPRC).

$$ACC = \frac{TP + TN}{TP + TN + FP + FN} \tag{20}$$

$$F1 = \frac{2 \cdot Precision \cdot Recall}{Precision + Recall} \tag{21}$$

$$Precision = \frac{TP}{TP + FP} \tag{22}$$

$$Recall = \frac{TP}{TP + FN} \tag{23}$$

where TP, TN, FP, and FN denote the number of true positives, true negatives, false positives, and false negatives, respectively.

D. Data Preprocessing

To ensure robust feature extraction and reliable classification performance, we performed a systematic preprocessing pipeline on all ECG datasets, addressing common artifacts and standardizing the data format.

Denoising. Raw ECG recordings typically contain various types of interference, such as noise, baseline wander, and motion artifacts, which adversely affect classification accuracy [43]. To mitigate these disturbances, we applied a Butterworth bandpass filter with cutoff frequencies of 0.05 and 75 Hz [44], preserving essential physiological information while reducing unwanted noise.

Downsampling. All ECG signals were downsampled from the original 500 Hz sampling rate to 100 Hz. This reduction significantly decreases computational complexity without substantial loss of diagnostic information, a practice consistent with previous studies in ECG-based modeling [45], [46].

Normalization. To alleviate potential distribution shift effects, instance normalization [47] was applied independently to each lead of every ECG record. This step ensures consistency in amplitude scales across different recordings, thereby improving model generalization [21].

Label Reconstruction. Original SNOMED-CT codes assigned to each ECG record were converted into discrete categorical labels. After this mapping process, the Ningbo, PTB-XL, and Chapman datasets contained 25, 22, and 19 distinct classes, respectively. Notably, all datasets exhibited varying degrees of class imbalance, characterized by substantial disparities in the distribution of positive versus negative samples within certain categories, as well as significant variation in the number of samples across different classes.

Through this preprocessing pipeline, we standardized input quality and format, laying a solid foundation for subsequent feature extraction and classification tasks.

E. Compared Methods

To comprehensively evaluate the effectiveness of our proposed GFM-MIP framework, we compare it against several representative self-supervised and contrastive learning baselines for time-series modeling.

1) **TF-C** [48]: A contrastive learning framework designed to extract temporally discriminative features from univariate and multivariate time series by contrasting representations of temporally shifted segments. It emphasizes capturing both global and local temporal dynamics.
2) **TS-TCC** [49]: This method constructs multiple augmented views of the same time series and applies contrastive objectives across different temporal perspectives. It enhances the model's robustness by explicitly learning temporal invariances.
3) **CPC** [50]: A predictive coding approach that maximizes mutual information between the current context and future latent representations. It encourages the encoder to retain predictive structure from the sequence, thus learning high-quality temporal representations.
4) **TimesURL** [38]: A unified pretraining framework that integrates multiple self-supervised objectives, such as

TABLE I
STATISTICS OF ECG DATASETS

Dataset	#Samples	Time-series	Laboratory Modality	Labels	Sampling frequency
Ningbo	34,905	12 leads, 10 s	✗	25 classes	500Hz
PTB-XL	21,837	12 leads, 10 s	✗	21 classes	500Hz
Chapman	10,247	12 leads, 10 s	✗	18 classes	500Hz
SDU-SH	2,123	12 leads, 10 s	✓	MI / Non-MI	500Hz

context prediction and sequence reordering. It aims to extract generalizable representations applicable to diverse downstream tasks.

5) **SimMTM** [51]: It learns temporal dependencies by randomly masking subsequences and reconstructing them, offering simplicity and effectiveness in pretraining.
6) **PatchTST** [21]: A pure Transformer-based forecasting model that partitions time series into non-overlapping patches. By utilizing global self-attention across patches, it achieves strong performance in capturing long-range temporal dependencies.
7) **TimeMAE** [37]: A masked autoencoding framework tailored for time-series data. It reconstructs randomly masked segments using encoder-decoder architecture and has shown competitive performance in transfer learning settings.

These methods serve as strong baselines for benchmarking multimodal and unimodal ECG representation learning.

F. Experimental Results

1) Performance Comparison on the SDU-SH Dataset with Complete Multimodal Inputs: To comprehensively evaluate the diagnostic performance of GFM-MIP in a practical clinical context, we conducted extensive experiments on the SDU-SH dataset, which comprises paired 12-lead ECG recordings and corresponding laboratory test results. For a thorough comparison, we selected various strong baseline methods, including supervised transformer variants (TF-C), temporal contrastive learning models (TS-TCC, CPC), self-supervised learning frameworks (TimesURL, SimMTM), and recent time-series transformer models (PatchTST, TimeMAE).

As shown in Table II, GFM-MIP demonstrated consistently superior performance across all four evaluation metrics—accuracy (ACC), F1 score, AUROC, and AUPRC—surpassing the baseline models. GFM-MIP outperformed all baselines on SDU-SH by +2.15% in AUROC and +2.33% in AUPRC, particularly excelling in high-sensitivity recall scenarios. Notably, GFM-MIP achieved significant improvements in the F1 score and AUPRC, indicating robust predictive performance and strong capabilities for addressing class imbalance. Compared with top-performing baselines such as TS-TCC, GFM-MIP exhibited clear advantages in AUROC and AUPRC, reflecting its superior ability to differentiate positive and negative clinical cases effectively.

While certain baselines, including TS-TCC and CPC, exhibited strong individual performances, they lacked explicit incorporation of patient-specific physiological information and multi-modal fusion mechanisms. By contrast, GFM-MIP leverages an early-stage FiLM modulation strategy that integrates

TABLE II
COMPARISON AMONG OURS MODEL AND BASELINE METHODS OVER THE SDU-SH DATASET.THE BEST RESULTS ARE BOLD AND THE SECOND BEST ARE UNDERLINED.

Models	SDU-SH			
	ACC	F1	AUROC	AUPRC
TF-C_1	93.73	89.42	95.93	92.26
TF-C_R	95.96	89.35	95.16	91.22
TS-TCC	<u>97.80</u>	<u>94.51</u>	<u>96.72</u>	95.59
CPC	95.44	90.70	96.45	<u>96.47</u>
TimesURL	96.43	93.18	95.75	94.48
SimMTM$_D$	92.20	88.60	94.35	85.34
SimMTM$_Q$	92.29	86.91	95.15	90.63
PatchTST	94.09	89.70	96.22	92.37
TimeMAE	95.22	92.71	96.53	95.61
GFM-MIP (Ours)	**98.82**	**96.36**	**98.87**	**98.80**

laboratory biomarkers, modality-specific encoders, and cross-modal alignment through contrastive learning. This sophisticated design enables GFM-MIP to effectively model patient heterogeneity and subtle diagnostic signals embedded within multimodal ECG data.

These findings highlight that GFM-MIP not only competes strongly against existing state-of-the-art methods but also demonstrates unique strengths when applied to rich, clinically annotated multimodal datasets. Its design facilitates precise and personalized predictions, emphasizing its practical clinical value.

2) Performance Comparison on Datasets without Laboratory Modality: To further validate GFM-MIP's generalizability under varied real-world conditions, we evaluated its performance on three publicly available ECG datasets: Ningbo, PTB-XL, and Chapman. Unlike the SDU-SH dataset, these public datasets do not include laboratory test features, necessitating a reduced variant of GFM-MIP without laboratory-based modulation. This scenario allowed us to investigate the framework's resilience when operating with incomplete modalities.

Even in the absence of laboratory data, Table III shows that GFM-MIP delivered strong and competitive results across all datasets. On the Ningbo dataset, GFM-MIP achieved the highest accuracy and F1 scores among the evaluated models, maintaining comparable AUROC scores relative to top-performing alternatives. Similarly, on the PTB-XL dataset, despite the lack of laboratory features, GFM-MIP exhibited consistently robust accuracy and balanced performance across metrics. On the Chapman dataset, GFM-MIP also achieved superior accuracy and F1 scores, demonstrating its reliable predictive capacity and generalization capability across datasets with varying distributions and label complexities.

Notably, several baseline models benefited from sophisticated self-supervised pretraining or specialized contrastive learning objectives. Nevertheless, even without complete multimodal input, GFM-MIP maintained strong performance through its fundamental architectural strengths, including modality-specific encoders and structured cross-modal interaction. These features allow GFM-MIP to extract robust and transferable representations from the available ECG modalities.

Overall, these experiments underscore the flexibility and robustness of GFM-MIP, highlighting its capability to maintain strong performance even when faced with modality limitations. This further supports its practical applicability and adaptability for diverse real-world ECG classification scenarios.

G. Ablation Experiments

To comprehensively evaluate the specific contributions of each component within the GFM-MIP framework, we systematically conducted ablation experiments shown in Table IV addressing both modality integration and structural design. These experiments provided detailed insights into how each element impacts the model's performance and robustness.

The model variants with specific modalities ablated are denoted as "w/o Laboratory," "w/o IMG," and "w/o TS," corresponding to the exclusion of laboratory test features, ECG images, and ECG time-series data, respectively. The complete version of our model is referred to as the "Full Model." In addition to modality ablations, we also investigate several architectural variations. The variant without contrastive learning is denoted as "w/o Contrast," while the model without the cross-modal fusion module is denoted as "w/o Fusion." Moreover, we examine two alternative fusion strategies: incorporating laboratory features only at the fusion stage, referred to as "Late-Concat," and employing a fully shared transformer fusion structure across all modalities, denoted as "All-Shared." These ablation settings are designed to isolate the contributions of each modality and architectural component to the overall performance of the GFM-MIP framework.

Initially, we investigated the role of incorporating multiple data modalities by individually removing them from the complete model. Eliminating the laboratory test features—which serve as personalized physiological context through FiLM modulation—resulted in noticeable performance degradation. This finding underscores the significance of integrating systemic biomarkers to enhance patient-specific modeling and improve classification accuracy. Similarly, when either ECG time series or ECG images were used exclusively, model performance significantly deteriorated compared to the multimodal baseline. Particularly, the absence of ECG time-series data markedly reduced performance, reflecting the critical role of electrophysiological dynamics, while the exclusion of ECG images confirmed the importance of morphological features. Thus, both temporal and morphological modalities offer unique and complementary diagnostic information that jointly enhances diagnostic effectiveness.

Furthermore, we examined the influence of key architectural elements by selectively removing or modifying them. Removing the contrastive learning objective negatively affected representation consistency and alignment across modalities, leading to less coherent multimodal representations and reduced classification capability. Omitting contrastive learning led to a 2.18% AUPRC drop, confirming its role in enforcing cross-modal alignment, as further illustrated in the latent space visualization (Fig. 2). Similarly, substituting the Transformer-based deep fusion module with a simpler concatenation strategy resulted in decreased model performance, highlighting the necessity of complex cross-modal interaction mechanisms for effectively capturing intricate modality interdependencies. Additionally, delaying the integration of laboratory features until after the fusion stage, rather than embedding them through early FiLM modulation, significantly compromised the generalization and diagnostic power of the model. This emphasizes the importance of early-stage personalization to effectively influence subsequent feature representation. Lastly, replacing modality-specific encoders with a shared generic encoder structure severely degraded performance, clearly indicating the necessity of distinct architectural designs tailored specifically for each data modality.

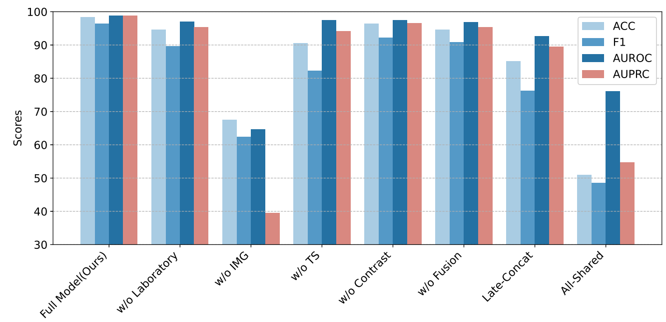

Fig. 2. Impact of modality and architectural ablations on classification metrics (SDU-SH dataset)

Collectively, these ablation studies shown in Table IV reveal that GFM-MIP's high diagnostic performance and robustness do not arise from any single isolated component. Instead, the model achieves its superior results through the cohesive integration of multimodal input data, personalized feature modulation via FiLM, sophisticated structured fusion through transformer-based modules, and representation alignment through contrastive learning. Each of these components contributes distinctly yet synergistically to the overall effectiveness, clinical interpretability, and generalization capability of the framework.

H. Parameter Analysis

1) Temperature Parameter and Contrastive Loss Weight: To investigate the influence of critical hyperparameters on model performance, we conduct a focused sensitivity analysis on two key components of our training objective: the temperature parameter τ in the contrastive loss and the weighting factor λ_{con} that balances the classification and contrastive objectives.

Temperature Parameter τ: As shown in Fig. 3, the model exhibits considerable sensitivity to the temperature scaling in the contrastive loss. Lower values (e.g., $\tau = 0.05$) tend to

TABLE III
COMPARISON AMONG OURS MODEL AND BASELINE METHODS ON PUBLIC DATASETS. THE BEST RESULTS ARE BOLD AND THE SECOND BEST ARE UNDERLINED.

Models	Ningbo				PTB-XL				Chapman			
	ACC	F1	AUROC	AUPRC	ACC	F1	AUROC	AUPRC	ACC	F1	AUROC	AUPRC
TF-C_I	55.87	75.30	92.83	49.09	51.08	78.84	91.06	<u>53.58</u>	59.59	77.10	<u>91.89</u>	<u>51.84</u>
TF-C_R	57.58	76.92	92.27	50.25	50.05	77.61	88.24	49.70	60.57	76.74	89.59	**52.27**
TS-TCC	<u>61.07</u>	<u>81.79</u>	92.61	53.72	<u>55.75</u>	**81.23**	88.42	50.60	<u>61.69</u>	<u>78.29</u>	89.18	46.28
CPC	59.26	79.32	**95.25**	**56.08**	55.24	<u>80.61</u>	**93.14**	**55.03**	59.25	75.97	**91.97**	46.33
TimesURL	59.43	81.20	<u>94.50</u>	53.49	53.85	80.04	<u>91.23</u>	50.99	55.44	75.96	89.99	48.13
$SimMTM_D$	54.85	75.67	90.79	44.68	52.98	78.84	87.17	45.03	45.88	61.12	86.43	38.50
$SimMTM_R$	55.02	74.10	91.48	49.36	54.67	80.43	85.80	47.24	59.44	75.74	88.25	47.03
PatchTST	55.28	77.05	93.47	50.07	54.90	80.40	90.64	50.05	53.34	73.93	91.21	49.28
TimeMAE	57.76	79.97	93.55	<u>54.30</u>	54.05	80.32	90.49	49.91	57.20	74.58	86.82	46.81
GFM-MIP (Ours)	**61.53**	**82.71**	91.91	52.81	**56.09**	79.40	88.58	51.36	**64.73**	**78.92**	89.84	48.78

TABLE IV
PERFORMANCE ANALYSIS OF OUR MODEL UNDER DIFFERENT ABLATION SETTINGS.

Variants	ACC	F1 Score	AUC	AUPRC
Full Model(Ours)	**98.82**	**96.36**	**98.87**	**98.80**
w/o Laboratory	94.58	89.59	96.99	95.35
w/o TS	67.53	62.32	64.63	39.48
w/o IMG	90.57	82.30	97.43	94.08
w/o Contrast	96.46	92.15	97.41	96.62
w/o Fusion	94.59	90.82	96.82	95.40
Late-Concat	85.14	76.23	92.62	89.42
All-Shared	50.94	48.51	76.02	54.77

produce sharper probability distributions, potentially leading to overfitting in representation alignment and performance degradation. Conversely, overly large values (e.g., $\tau = 1.0$) flatten the similarity scores, reducing the discriminative power of contrastive pairs. The model achieves optimal F1 performance when τ lies in the intermediate range of approximately 0.1 to 0.2, suggesting a balanced trade-off between contrast sharpness and generalization.

Contrastive Loss Weight λ_{con}: Varying the contribution of the contrastive loss reveals its importance in representation learning. With no contrastive component ($\lambda_{\text{con}} = 0$), the model relies solely on supervised classification, which leads to underutilization of cross-modal alignment. As λ_{con} increases, the model benefits from enhanced modality interaction, yielding improved performance up to a point. However, excessively high values (e.g., $\lambda_{\text{con}} = 1.0$) begin to suppress the supervised signal, causing a slight decline in performance. These observations validate the necessity of balancing contrastive and classification signals, with the best performance achieved when λ_{con} is set between 0.3 and 0.5.

This analysis highlights the critical role of careful hyperparameter tuning in contrastive learning-based multimodal frameworks. Optimal values of τ and λ_{con} not only improve classification performance but also enhance the robustness of modality alignment, ultimately leading to more generalizable and reliable clinical predictions.

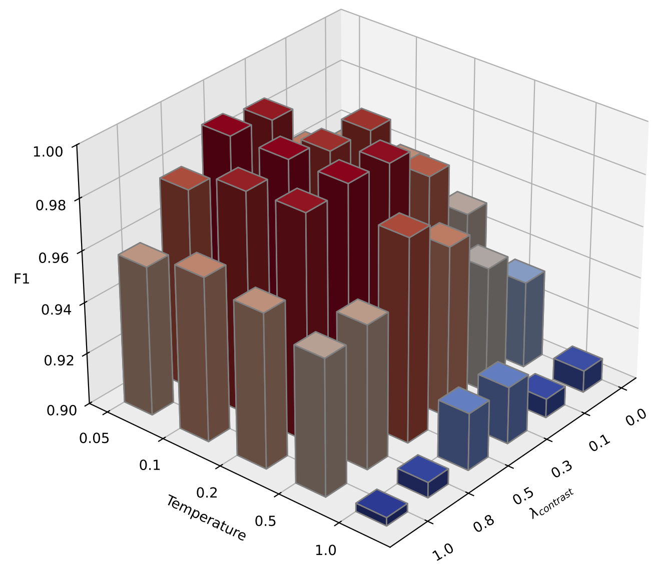

Fig. 3. The model performance with different parameter settings over SDU-SH Dataset

2) Fusion Layer Depth: To evaluate how the number of fusion layers affects model performance, we conducted an ablation analysis by varying the number of Transformer layers in the cross-modal fusion module from 2 to 8. As shown in Fig. 4, performance trends are consistent across all four datasets: SDU-SH, Ningbo, PTB-XL, and Chapman. Specifically, model accuracy improves with increasing fusion depth up to 6 layers, after which the performance either plateaus or slightly declines.

This suggests that a moderate depth offers the best trade-off between representational richness and overfitting risk. A shallow fusion module may fail to capture complex cross-modal interactions, while excessive depth can introduce noise or lead to optimization difficulties. These findings highlight the importance of tuning the fusion depth to balance expressiveness and generalization in multimodal architectures like

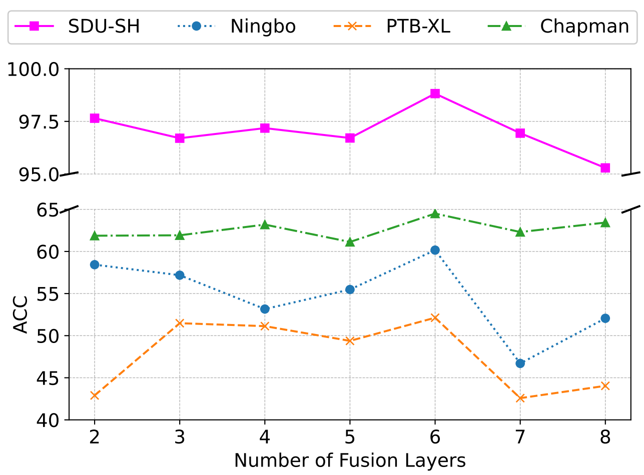

Fig. 4. Impact of fusion depth on classification accuracy across datasets

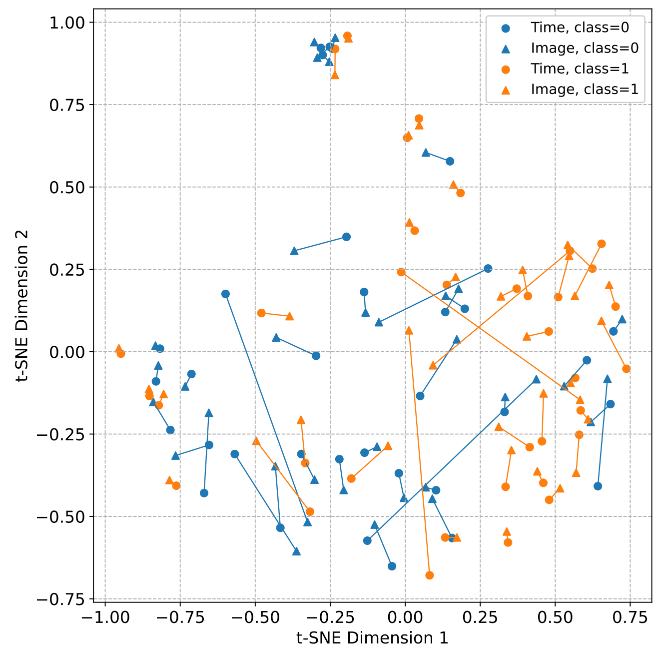

Fig. 5. Latent Space Visualization of Modality Alignment via Contrastive Learning. Connected pairs represent same patients.

GFM-MIP.

I. Visualization Analysis

To further understand the impact of contrastive learning on cross-modal representation alignment, we conduct a qualitative visualization of the learned feature embeddings from both the ECG time-series and image branches. Specifically, we project the high-dimensional modality-specific embeddings into a two-dimensional space using t-SNE. For each sample, we plot its ECG time-series embedding ($\mathbf{z}_{\text{ts}}$) and its corresponding image embedding ($\mathbf{z}_{\text{img}}$) as paired points, connected by a dashed line to represent the same patient.

As illustrated in Figure 5, GFM-MIP effectively aligns representations across modalities. The embeddings of the same sample from different modalities are positioned closely in the latent space, demonstrating successful cross-modal feature consistency. This alignment is a direct consequence of the bidirectional contrastive loss, which encourages paired samples to share semantic similarity despite modality heterogeneity.

In addition to pairwise proximity, the scatter plot also reveals clear clustering patterns based on diagnostic labels. Samples from different classes tend to form distinct clusters, indicating that the shared embedding space preserves discriminative information for classification. This separation highlights the model's ability to encode label-relevant structure while maintaining cross-modal coherence.

These observations confirm the dual benefit of our contrastive learning strategy: it not only improves intra-sample alignment across modalities but also enhances inter-class separability, contributing to both robust fusion and accurate prediction in the GFM-MIP framework.

Further inspection of incorrectly classified instances reveals that these points tend to exhibit weaker alignment between modalities, underscoring the practical importance of our contrastive learning objective. Such insights suggest potential directions for targeted model refinement or data augmentation.

V. Discussion

GFM-MIP integrates 12-lead ECG time-series, ECG images, and laboratory biomarkers within a single framework and achieves strong performance on our clinical cohort. These gains stem from the combination of structured temporal modeling of inter-lead relations via a Graphormer encoder, early patient-conditioned FiLM modulation driven by laboratory features, a Transformer-based deep fusion module that captures cross-modal interactions, and a symmetric contrastive objective that aligns modality-specific representations.

Methodologically, the framework advances prior art along several axes. Laboratory biomarkers are treated as primary sources of information and injected early via FiLM to personalize representation learning; the Graphormer explicitly encodes inter-lead dependencies rather than treating leads as independent channels; and contrastive learning regularizes the shared embedding space to enhance semantic coherence across modalities. Ablation studies (Table 4) indicate that removing FiLM, contrastive alignment, or deep fusion consistently degrades performance, underscoring their complementary—not redundant—contributions.

On Ningbo, PTB-XL, and Chapman (Table 3), GFM-MIP is competitive but not uniformly superior. This is expected because these benchmarks lack laboratory features—the modality that our framework uniquely leverages and from which it derives its largest benefit. When restricted to the available modalities, the model maintains robust accuracy and F1; however, the advantage of early, patient-specific FiLM conditioning cannot fully manifest. By contrast, on our clinical dataset—where laboratory data are available—GFM-MIP attains the best results across all metrics (Table 2), indicating that the proposed innovation is most impactful in genuinely multimodal clinical settings.

The design choices align closely with clinical reasoning. Electrophysiological patterns (signals), morphological structures (images), and biochemical indicators (laboratory tests) provide complementary evidence for myocardial infarction. FiLM translates laboratory context into feature-wise scaling and shifting, analogous to how clinicians weigh troponin or inflammatory markers when interpreting borderline ECGs. Contrastive alignment improves consistency between signal- and image-derived evidence, which is particularly helpful in ambiguous cases; latent-space visualizations show tighter cross-modal pairing for correctly classified samples.

Analysis of model behavior yields practical guidance. Deep fusion confers substantial gains over simple concatenation, with diminishing returns beyond moderate depth; empirically, approximately six fusion layers balance expressiveness and generalization (Fig. 4). Sensitivity analyses (Fig. 3) further suggest that intermediate contrastive temperatures and modest contrastive weights provide favorable trade-offs between alignment and the supervised signal, informing deployment and hyperparameter selection in clinical pipelines.

There are several limitations in our current work. Firstly, laboratory features are not consistently available across clinical settings or public benchmarks, which limits the benefit of our FiLM-based personalization. When labs are missing, the model cannot exploit patient-specific biochemical context. To mitigate this, we plan to adopt training strategies that are robust to missing modalities, such as explicit missingness modeling, conditional or gated FiLM, and generative or multiple-imputation schemes. We also aim to produce uncertainty-aware outputs that flag elevated risk under incomplete inputs, and to pursue prospective data collection that standardizes the capture of key biomarkers. Second, retrospective labels may be noisy, and differences in population, devices, and clinical workflows can induce distribution shift, potentially undermining external validity. We will address this through harmonization of acquisition and preprocessing, robust learning methods for noisy labels, including label correction and co-teaching, and domain generalization or adaptation to handle site-specific shifts. After deployment, shift detection with recalibration—such as temperature scaling and class-conditional thresholds—together with continuous performance monitoring, will be essential. Additionally, the depth of the fusion module and the use of dual encoders increase parameters, FLOPs, and memory footprint, which can hinder deployment in resource-constrained environments. We will pursue efficiency via structured pruning and low-rank factorization, quantization-aware training, knowledge distillation to lightweight students or shallower fusion backbones, and dynamic inference with token or patch pruning and early-exit mechanisms. Operator- and graph-level optimizations, paired with reliability calibration, will be evaluated alongside strict latency, throughput, and energy benchmarks to ensure clinically viable performance. Addressing these limitations should strengthen robustness, generalizability, and practical feasibility.

Taken together, unifying structured temporal modeling, laboratory-conditioned personalization, deep cross-modal fusion, and contrastive alignment meaningfully advances ECG-based MI prediction. GFM-MIP not only improves diagnostic performance but also delineates a clinically actionable path toward patient-specific diagnostic tools—especially in settings where laboratory biomarkers can be incorporated early in the inference process.

VI. Conclusion

In this study, we introduced GFM-MIP, a comprehensive multimodal framework designed for effective and interpretable ECG-based cardiovascular disease prediction. By combining ECG time-series data, ECG images, and laboratory test results, GFM-MIP addresses the limitations inherent in single-modality approaches, providing a more holistic and clinically meaningful representation of patient health.

GFM-MIP integrates several innovative design elements. Modality-specific encoders were employed to accurately capture the distinct features of time-series signals and images. Additionally, we introduced Feature-wise Linear Modulation (FiLM) to embed patient-specific physiological information derived from laboratory tests into the encoding process, facilitating personalized representation learning. A Transformer-based fusion module was also incorporated to effectively model the rich interactions between different modalities, while contrastive learning further strengthened the semantic alignment across these diverse data sources.

Extensive experimental evaluations clearly demonstrate that GFM-MIP achieves superior performance compared to various baseline models and ablation variants across multiple metrics. These results underline the importance of multimodal integration, early-stage physiological conditioning, and structured cross-modal fusion in improving diagnostic accuracy and enhancing model robustness.

Future research could explore several promising directions. Incorporating additional modalities, such as clinical notes, echocardiographic imaging, or genomic data, could further enrich the patient representations. Extending the framework to multi-label or multi-task scenarios would broaden its clinical applicability, potentially supporting more complex diagnostic tasks. Investigations into the interpretability of learned representations and real-world clinical implementation will also be valuable for enhancing clinical adoption.

In summary, this study demonstrates that structured fusion of ECG modalities and laboratory features—guided by alignment loss and patient context—can meaningfully enhance diagnostic precision in real-world cardiovascular applications.

[1] S. Hong, Y. Zhou, J. Shang *et al.*, "Opportunities and challenges of deep learning methods for electrocardiogram data: A systematic review," *Computers in Biology and Medicine*, vol. 122, p. 103801, Jul. 2020.

[2] S. Sahoo, M. Dash, S. Behera *et al.*, "Machine learning approach to detect cardiac arrhythmias in ECG signals: A survey," *Irbm*, vol. 41, no. 4, pp. 185–194, Aug. 2020.

[3] A. Esteva, A. Robicquet, B. Ramsundar *et al.*, "A guide to deep learning in healthcare," *Nature Medicine*, vol. 25, no. 1, pp. 24–29, Jan. 2019.

[4] O. Attallah and D. A. Ragab, "Auto-MyIn: Automatic diagnosis of myocardial infarction via multiple GLCMs, CNNs, and SVMs," *Biomedical Signal Processing and Control*, vol. 80, p. 104273, Feb. 2023.

[5] J. Hassannataj Joloudari, S. Mojrian, I. Nodehi *et al.*, “Application of artificial intelligence techniques for automated detection of myocardial infarction: A review,” *Physiological Measurement*, vol. 43, no. 8, p. 8TR01, Aug. 2022.

[6] M. Jafari, A. Shoeibi, M. Khodatars *et al.*, “Automated diagnosis of cardiovascular diseases from cardiac magnetic resonance imaging using deep learning models: A review,” *Computers in Biology and Medicine*, vol. 160, p. 106998, Jun. 2023.

[7] V. Jahmunah, E. Ng, T. R. San *et al.*, “Automated detection of coronary artery disease, myocardial infarction and congestive heart failure using GaborCNN model with ECG signals,” *Computers in Biology and Medicine*, vol. 134, p. 104457, Jul. 2021.

[8] J. T. Neumann, R. Twerenbold, F. Ojeda *et al.*, “Personalized diagnosis in suspected myocardial infarction,” *Clinical Research in Cardiology*, vol. 112, no. 9, pp. 1288–1301, Sep. 2023.

[9] A. Schuster, T. Lange, S. J. Backhaus *et al.*, “Fully automated cardiac assessment for diagnostic and prognostic stratification following myocardial infarction,” *Journal of the American Heart Association*, vol. 9, no. 18, p. e016612, Sep. 2020.

[10] M. Sharma, R. S. Tan, and U. R. Acharya, “A novel automated diagnostic system for classification of myocardial infarction ECG signals using an optimal biorthogonal filter bank,” *Computers in Biology and Medicine*, vol. 102, pp. 341–356, Nov. 2018.

[11] M. Tsipouras, T. Exarchos, D. Fotiadis *et al.*, “Automated diagnosis of coronary artery disease based on data mining and fuzzy modeling,” *IEEE Transactions on Information Technology in Biomedicine*, vol. 12, no. 4, pp. 447–458, Jul. 2008.

[12] J. Zhang, F. Lin, P. Xiong *et al.*, “Automated detection and localization of myocardial infarction with staked sparse autoencoder and TreeBagger,” *IEEE Access*, vol. 7, pp. 70 634–70 642, 2019.

[13] B. Toprak, H. Solleder, E. Di Carluccio *et al.*, “Diagnostic accuracy of a machine learning algorithm using point-of-care high-sensitivity cardiac troponin I for rapid rule-out of myocardial infarction: A retrospective study,” *The Lancet Digital Health*, vol. 6, no. 10, pp. e729–e738, Oct. 2024.

[14] Z. H. Kilimci, M. Yalcin, A. Kucukmanisa *et al.*, “Heart disease detection using vision-based transformer models from ECG images.”

[15] J. Tang, T. Xia, Y. Lu *et al.*, “Electrocardiogram-language model for few-shot question answering with meta learning,” Oct. 2024.

[16] L. Holmstrom, M. Christensen, N. Yuan *et al.*, “Deep learning-based electrocardiographic screening for chronic kidney disease,” *Communications Medicine*, vol. 3, no. 1, p. 73, May 2023.

[17] C. Yu, X. Ma, J. Ren *et al.*, “Spatio-temporal graph transformer networks for pedestrian trajectory prediction,” in *Computer Vision–ECCV 2020: 16th European Conference, Glasgow, UK, August 23–28, 2020, Proceedings, Part XII 16*. Springer, 2020, pp. 507–523.

[18] W.-C. Liu, C.-S. Lin, C.-S. Tsai *et al.*, “A deep learning algorithm for detecting acute myocardial infarction,” *Eurointervention*, vol. 17, no. 9, pp. 765–773, Oct. 2021.

[19] K. Jafarian, V. Vahdat, S. Salehi *et al.*, “Automating detection and localization of myocardial infarction using shallow and end-to-end deep neural networks,” *Applied Soft Computing*, vol. 93, p. 106383, Aug. 2020.

[20] C.-H. Lee, W.-T. Liu, Y.-S. Lou *et al.*, “Artificial intelligence-enabled electrocardiogram screens low left ventricular ejection fraction with a degree of confidence,” *DIGITAL HEALTH*, vol. 8, p. 205520762211432, Jan. 2022.

[21] Y. Nie, N. H. Nguyen, P. Sinthong *et al.*, “A time series is worth 64 words: Long-term forecasting with transformers,” Mar. 2023.

[22] S. Khurshid, S. Friedman, C. Reeder *et al.*, “ECG-based deep learning and clinical risk factors to predict atrial fibrillation,” *Circulation*, vol. 145, no. 2, pp. 122–133, Jan. 2022.

[23] N. Yuan, G. Duffy, S. S. Dhruva *et al.*, “Deep learning of electrocardiograms in sinus rhythm from US veterans to predict atrial fibrillation,” *JAMA Cardiology*, vol. 8, no. 12, p. 1131, Dec. 2023.

[24] S. J. Backhaus, H. Aldehayat, J. T. Kowallick *et al.*, “Artificial intelligence fully automated myocardial strain quantification for risk stratification following acute myocardial infarction,” *Scientific Reports*, vol. 12, no. 1, p. 12220, Jul. 2022.

[25] J.-H. Nam, N. S. Syazwany, and S.-C. Lee, “Transgunet: Transformer meets graph-based skip connection for medical image segmentation,” *arXiv preprint arXiv:2502.09931*, 2025.

[26] B. Li and S. Nabavi, “A multimodal graph neural network framework for cancer molecular subtype classification,” *BMC bioinformatics*, vol. 25, no. 1, p. 27, 2024.

[27] X. Li, Y. Zhou, N. Dvornek *et al.*, “Braingnn: Interpretable brain graph neural network for fmri analysis,” *Medical Image Analysis*, vol. 74, p. 102233, 2021.

[28] J. Lampert, A. Vaid, W. Whang *et al.*, “A novel ECG-based deep learning algorithm to predict cardiomyopathy in patients with premature ventricular complexes,” *JACC: Clinical Electrophysiology*, vol. 9, no. 8, pp. 1437–1451, Aug. 2023.

[29] J. Boeddinghaus, D. Doudesis, P. Lopez-Ayala *et al.*, “Machine learning for myocardial infarction compared with guideline-recommended diagnostic pathways,” *Circulation*, vol. 149, no. 14, pp. 1090–1101, Apr. 2024.

[30] D. Doudesis, K. K. Lee, J. Boeddinghaus *et al.*, “Machine learning for diagnosis of myocardial infarction using cardiac troponin concentrations,” *Nature Medicine*, vol. 29, no. 5, pp. 1201–1210, May 2023.

[31] S. Al-Zaiti, C. Martin-Gill, J. Zègre-Hemsey *et al.*, “Machine learning for the ECG diagnosis and risk stratification of occlusion myocardial infarction at first medical contact,” Jan. 2023.

[32] S. S. Al-Zaiti, C. Martin-Gill, J. K. Zègre-Hemsey *et al.*, “Machine learning for ECG diagnosis and risk stratification of occlusion myocardial infarction,” *Nature Medicine*, vol. 29, no. 7, pp. 1804–1813, Jul. 2023.

[33] W. Sun, S. V. Kalmady, N. Sepehrvand *et al.*, “Towards artificial intelligence-based learning health system for population-level mortality prediction using electrocardiograms,” *npj Digital Medicine*, vol. 6, no. 1, p. 21, Feb. 2023.

[34] S. V. Kalmady, A. Salimi, W. Sun *et al.*, “Development and validation of machine learning algorithms based on electrocardiograms for cardiovascular diagnoses at the population level,” *npj Digital Medicine*, vol. 7, no. 1, p. 133, May 2024.

[35] G. N. Kowlgi, Z. I. Attia, and S. J. Asirvatham, “Deep learning for premature ventricular contraction-cardiomyopathy,” *JACC: Clinical Electrophysiology*, vol. 9, no. 8, pp. 1452–1454, Aug. 2023.

[36] Y. Wei, C. Lian, B. Xu *et al.*, “Bimodal Masked Autoencoders with internal representation connections for electrocardiogram classification,” *Pattern Recognition*, vol. 161, p. 111311, May 2025.

[37] M. Cheng, Q. Liu, Z. Liu *et al.*, “TimeMAE: Self-supervised representations of time series with decoupled masked autoencoders,” Mar. 2023.

[38] J. Liu and S. Chen, “TimesURL: Self-supervised contrastive learning for universal time series representation learning,” *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 38, no. 12, pp. 13 918–13 926, Mar. 2024.

[39] J. Zheng, H. Chu, D. Struppa *et al.*, “Optimal multi-stage arrhythmia classification approach,” *Scientific Reports*, vol. 10, no. 1, p. 2898, Feb. 2020.

[40] P. Wagner, N. Strodthoff, R.-D. Bousseljot *et al.*, “PTB-XL, a large publicly available electrocardiography dataset,” *Scientific Data*, vol. 7, no. 1, p. 154, May 2020.

[41] J. Zheng, J. Zhang, S. Danioko *et al.*, “A 12-lead electrocardiogram database for arrhythmia research covering more than 10,000 patients,” *Scientific Data*, vol. 7, no. 1, p. 48, Feb. 2020.

[42] M. A. Reyna, N. Sadr, E. A. P. Alday *et al.*, “Will two do? Varying dimensions in electrocardiography: The PhysioNet/computing in cardiology challenge 2021,” in *2021 Computing in Cardiology (Cinc)*. Brno, Czech Republic: IEEE, Sep. 2021, pp. 1–4.

[43] X. Xie, H. Liu, M. Shu *et al.*, “A multi-stage denoising framework for ambulatory ecg signal based on domain knowledge and motion artifact detection,” *Future Generation Computer Systems*, vol. 116, pp. 103–116, 2021.

[44] V. Krasteva, I. Jekova, R. Leber *et al.*, “Real-time arrhythmia detection with supplementary ecg quality and pulse wave monitoring for the reduction of false alarms in icus,” *Physiological measurement*, vol. 37, no. 8, p. 1273, 2016.

[45] H. Zhang, W. Liu, J. Shi *et al.*, “MaeFE: Masked autoencoders family of electrocardiogram for self-supervised pretraining and transfer learning,” *IEEE Transactions on Instrumentation and Measurement*, vol. 72, pp. 1–15, 2023.

[46] H. Martin, U. Morar, W. Izquierdo *et al.*, “Real-time frequency-independent single-lead and single-beat myocardial infarction detection,” *Artificial intelligence in medicine*, vol. 121, p. 102179, 2021.

[47] D. Ulyanov, A. Vedaldi, and V. Lempitsky, “Instance normalization: The missing ingredient for fast stylization,” Nov. 2017.

[48] X. Zhang, Z. Zhao, T. Tsiligkaridis *et al.*, “Self-supervised contrastive pre-training for time series via time-frequency consistency.”

[49] E. Eldele, M. Ragab, Z. Chen *et al.*, “Time-series representation learning via temporal and contextual contrasting,” Jun. 2021.

[50] T. Mehari and N. Strodthoff, “Self-supervised representation learning from 12-lead ECG data,” *Computers in Biology and Medicine*, vol. 141, p. 105114, Feb. 2022.
[51] J. Dong, H. Wu, H. Zhang *et al.*, “SimMTM: A simple pre-training framework for masked time-series modeling.”